Assignment-4

Take one domain and build business Understanding

Business Understanding Phase – Data Science Project

<u>Domain Chosen: Online Retail (E-Commerce Platform)</u>

1. Introduction to the Domain

Online retail, also known as e-commerce, refers to the buying and selling of goods and services over the internet. Major global platforms such as Amazon, Flipkart, and eBay operate in this domain. These platforms handle millions of customers, products, transactions, reviews, and logistics operations daily.

Because of the large volume of data generated (customer clicks, purchases, ratings, search history, payment methods, etc.), online retail is an ideal domain for data science applications. Data science helps businesses improve sales, customer satisfaction, personalization, inventory management, and marketing strategies.

The Business Understanding phase is the first and most critical step in a data science project. It focuses on clearly defining business goals, identifying problems, understanding constraints, and determining how data-driven solutions can create value.

2. Business Problem Identification

In the online retail domain, one major problem is:

Problem Statement:

Low customer retention and decreasing repeat purchase rate.

Many customers visit the platform, make one purchase, and never return. This leads to:

- High customer acquisition cost
- Reduced long-term revenue
- Poor customer lifetime value (CLV)
- Inefficient marketing spend

Another common issue includes:

- Cart abandonment
- Poor product recommendations
- Overstock or understock of products
- Ineffective personalized marketing

For this project, we focus on:

Objective: Improve customer retention by predicting churn and recommending personalized products.

3. Business Objectives

The primary business objectives are:

1. Increase repeat purchases by at least 15%.
2. Reduce customer churn rate.
3. Improve customer lifetime value (CLV).
4. Increase conversion rate through better recommendations.
5. Optimize marketing campaigns to target the right customers.

These objectives should align with overall company goals such as:

- Revenue growth
- Profit maximization
- Competitive advantage

- Customer satisfaction improvement

4. Business Success Criteria

To evaluate success from a business perspective, the following measurable criteria are defined:

- Increase repeat purchase rate from 30% to 45%.
- Reduce churn rate by 10%.
- Improve email campaign click-through rate (CTR) by 20%.
- Increase average order value (AOV).

If these targets are achieved, the project will be considered successful from a business standpoint.

5. Data Science Objectives

The business goals must be translated into technical data science goals:

1. Build a customer churn prediction model.
2. Develop a recommendation system.
3. Perform customer segmentation.
4. Predict customer lifetime value (CLV).

6. Data Mining Goals

To achieve the above objectives, we define specific data mining tasks:

- Classification: Predict whether a customer will churn or not.
- Clustering: Group customers into segments based on behavior.
- Recommendation: Suggest relevant products.
- Regression: Predict future purchase value.

7. Stakeholders Involved

The key stakeholders in this project include:

1. Business Managers – Define business goals.

2. Marketing Team – Use insights for campaigns.
3. Product Team – Improve user experience.
4. Data Science Team – Build and deploy models.
5. IT/Engineering Team – Implement models in production.
6. Customers – End users benefiting from personalization.

Each stakeholder has different expectations, so alignment is essential.

8. Business Constraints

Several constraints may affect the project:

- Budget limitations
- Limited data quality
- Data privacy regulations
- Time constraints
- Technology infrastructure limitations

For example, platforms operating in India must comply with local IT and data protection regulations.

9. Assumptions

The project assumes:

- Historical customer data is available.
- Data is clean and properly stored.
- Customer behavior patterns are predictable.
- Sufficient computing infrastructure exists.

If these assumptions are incorrect, the project may face challenges.

10. Risks Assessment

Potential risks include:

- Poor data quality leading to inaccurate predictions.
- Biased recommendation system.

- Overfitting models.
- Customer privacy concerns.
- Negative user experience if recommendations are irrelevant.

Risk mitigation strategies should be planned in advance.

11. Cost-Benefit Analysis

Costs:

- Data scientist salaries
- Infrastructure costs
- Software tools
- Development and deployment expenses

Benefits:

- Increased revenue
- Better customer engagement
- Reduced marketing waste
- Improved brand loyalty

If the projected revenue increase exceeds implementation costs, the project is financially viable.

12. Project Timeline (High-Level)

1. Business Understanding – 1–2 weeks
2. Data Collection & Preparation – 3–4 weeks
3. Model Development – 3 weeks
4. Testing & Evaluation – 2 weeks
5. Deployment – 2 weeks
6. Monitoring & Maintenance – Ongoing

13. Expected Business Impact

If successfully implemented, the system will:

- Increase customer retention.
- Improve personalization experience.
- Reduce churn.
- Enhance revenue growth.
- Strengthen competitive positioning in the e-commerce market.

Over time, data-driven decision-making will become part of the company's culture, improving operational efficiency and strategic planning.

14. Conclusion

The Business Understanding phase is the foundation of any successful data science project. In the online retail (e-commerce) domain, improving customer retention and personalization can significantly impact revenue and long-term growth.

By clearly defining business objectives, identifying stakeholders, assessing risks, and translating business problems into data science goals, the organization can ensure that the project delivers measurable business value.

A well-defined Business Understanding phase reduces ambiguity, aligns technical teams with business goals, and increases the chances of successful project implementation in a competitive e-commerce environment.